MATH328 Reference

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1 Linear Algebra

- The **zero matrix** is a matrix of all zeros.
- The **identity matrix** I_n is a square matrix of all ones on the main diagonal and zeros elsewhere.
- **Matrix multiplication**: The product of two matrices A and B is defined as
$$AB = \left(\sum_{k=1}^{n} a_{ik} b_{kj}\right)$$
Note: the inner dimensions must match for matrix multiplication to be defined. Matrix multiplication is *not* commutative, i.e., $AB \neq BA$ n general. The **commutator** of two matrices A and B is defined as
$$[A, B] = AB - BA$$
If $[A, B] = 0$, then A and B **commute**.
- **Linear functions and operators**: A function or operator $T(\mathbf{x}) = A\mathbf{x}$ is linear iff
  - $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ for any vectors $\mathbf{x}$ and $\mathbf{y}$, and
  - $T(c\mathbf{x}) = cT(\mathbf{x})$ for any scalar c and vector $\mathbf{x}$.
- **Elementary row operations**:
  - Swap two rows
  - Multiply a row by a nonzero scalar
  - Add a multiple of one row to another row

  These help solve matrix equations such as $A\mathbf{x} = \mathbf{b}$.
- The **rank** of a matrix is the number of non-zero rows in the echelon form of a matrix.
- Consistency and uniqueness of solutions of a matrix equation $A\mathbf{x} = \mathbf{b}$, where n is the number of columns in A:

- $\text{rank}(A) < \text{rank}(A \mid b) \implies$ **inconsistent**.
- $\text{rank}(A) = \text{rank}(A \mid b) = n \implies$ **unique solution**.
- $\text{rank}(A) = \text{rank}(A \mid b) = R < n \implies$ **infinitely many solutions**, with $n - R$ free variables.

- A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is **linearly independent** if the only solution to the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

is $c_1 = c_2 = \cdots = c_n = 0$. In other words,

$$\text{rank}\begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n\end{pmatrix} = n,$$

meaning

$$A\mathbf{x} = \mathbf{0}$$

has only the trivial solution $\mathbf{x} = \mathbf{0}$.

When vectors contained in a set are placed as columns into a matrix A, $\text{rank}(A)$ is the dimension of the vector space spanned by the vectors.

- **Computing determinants**:
  - $n = 1$: $A = \begin{pmatrix}a\end{pmatrix}$, $\det A = a$
  - $n = 2$: $A = \begin{pmatrix}a & b \\ c & d\end{pmatrix}$, $\det A = ad - bc$
  - $n \geq 3$: The determinant can be computed by expanding the minor determinants (M_{ij}) of the elements (a_{ij}) of *any* row or column:

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(M_{ij})$$

- **Properties of determinants**:
  - $k \det A = \det B$ if a row or column of B is k times the corresponding row or column of A.
  - $\det(kA) = k^n \det A$ if A is an $n \times n$ matrix.
  - $detA = 0$ if
    - A has a row or column of zeros,
    - A has a row or column that is a multiple of another row or column, or
    - A is not invertible.
  - $\det B = (-1)^r \det A$ if B is obtained from A by swapping two rows or columns r times.
  - $\det(A^T) = \det(A)$

- $\det A = \det U$ where U is the upper triangular matrix obtained from A by elementary row operations. $\det U$ is also the product of the diagonal entries of U.
- $\det(AB) = \det(A)\det(B)$

- **Linear independence of functions**: A set of functions $\{f_0, f_1, \ldots, f_n\}$ is linearly independent if the Wronstian determinant is non-zero:

$$W(f_0, f_1, \ldots, f_n) = \begin{vmatrix} f_0 & f_1 & \cdots & f_n \\ f_0' & f_1' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_0^{(n)} & f_1^{(n)} & \cdots & f_n^{(n)} \end{vmatrix} \neq 0$$

- **Cramer's Rule**: For a system of linear equations $A\mathbf{x} = \mathbf{b}$, the solution for x_i is given by

$$x_i = \frac{\det A_i}{\det A}$$

  where A_i is the matrix obtained from A by replacing the i-th column with $\mathbf{b}$. Note that this only works when $\det A \neq 0$.

- The **matrix inverse** A^{-1} of A is such that $AA^{-1} = A^{-1}A = I$. It can be computed via

$$A^{-1} = \frac{1}{\det A} C^T,$$

  where C is the matrix of cofactors of A. Remember that to find the cofactor C_{ij}, you need to find the determinant of the matrix A with the i-th row and j-th column removed, and then multiply it by $(-1)^{i+j}$.

  The matrix inverse can also be computed by augmenting the matrix A with the identity matrix I and performing elementary row operations until the left side is the identity matrix. The right side will then be the inverse of A.

  Not all matrices are invertible. A matrix is **invertible** iff its determinant is non-zero. In other words, A is invertible iff $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.

- **Eigenvalues and Eigenvectors**: For an $n \times n$ matrix A, an eigenvector $\mathbf{v}$ and its corresponding eigenvalue λ satisfy the equation

$$A\mathbf{v} = \lambda\mathbf{v},$$

  where $\mathbf{v} \neq \mathbf{0}$. The eigenvalues are the roots of the characteristic equation

$$\det(A - \lambda I) = 0.$$

  The eigenvectors are the non-zero solutions to

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

There may be up to n eigenvalues (including multiplicity) and up to n corresponding eigenvectors. Eigenvectors may be scaled by any non-zero scalar.

- **Similarity transformation**: A square matrix A can be converted to a diagonal matrix D via a similarity transformation

$$D = C^{-1}AC,$$

where C is the matrix of eigenvectors and D is the diagonal matrix of eigenvalues. We say that C *diagonalizes* A or that A is **similar** to D. This is because D describes the same transformation as A but in a new coordinate system.

- An **orthogonal matrix** Q is a square matrix such that $Q^TQ = QQ^T = I$. The columns (and rows) of an orthogonal matrix form an orthonormal basis, i.e., they are orthogonal to each other and have unit length. The action of Q on a vector preserves its length.

  Note that $\det Q = \pm 1$. In particular:

  - If $\det Q = 1$, then Q is a rotation matrix.
  - If $\det Q = -1$, then Q is a reflection matrix.

- If a square matrix A is symmetric, its eigenvectors are **orthogonal**. After normalizing them, the matrix C of orthonormal eigenvectors is an **orthogonal matrix**. The similarity transformation then becomes

$$D = C^{-1}AC = C^TAC,$$

which can be used to define a new *rotated orthogonal* coordinate system.

- A **rotation matrix** $R(\theta)$ performs a counter-clockwise rotation by angle θ and is given by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Note that $R(\theta)$ is an **orthogonal matrix**. Also note that $R(\theta)^T$ represents a clockwise rotation by angle θ.

Applying $R(\theta)$ to the standard basis vectors yields the rotated basis vectors.

An example of a rotation matrix in 3D is

$$R(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}.$$

Note that $R(\theta)$ is a passive transformation (corresponding to the $-\sin\theta$ in the bottom-left entry) about the y-axis (corresponding to the 2nd row matching I_3).

- A transformation is an **active transformation** if it transforms the *vector* $\mathbf{x}$ to a new vector $\mathbf{y}$ in the *same* coordinate system. For example, A below rotates the vector $\mathbf{x}$ counter-clockwise by angle θ to $\mathbf{y}$:

$$\mathbf{y} = A\mathbf{x}$$
$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

- A transformation is a **passive transformation** if it transforms the *coordinate system* (e.g., from unprimed to primed) *without* changing the vector $\mathbf{x}$ (i.e., $\mathbf{x} = \mathbf{x}'$). For example, A below rotates the coordinate system counter-clockwise by angle θ:

$$\mathbf{x}' = A\mathbf{x}$$
$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

- **Central conic equation in matrix form**:

$$Ax^2 + 2Hxy + By^2 = K$$
$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & H \\ H & B \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = K$$
$$\mathbf{x}^T M \mathbf{x} = K$$

- **Quadric surface equation in matrix form**:

$$Ax^2 + By^2 + Cz^2 + 2Dxy + 2Exz + 2Fyz = K$$
$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} A & D & E \\ D & B & F \\ E & F & C \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = K$$
$$\mathbf{x}^T M \mathbf{x} = K$$

  Instead of memorizing the formulas, remember that the matrix is symmetric and the diagonal entries are the coefficients of the squared terms, and the off-diagonal entries are half the coefficients of the cross terms. Also, the indices of the cross terms match the variables. For example, the term a_{12} corresponds to the cross term xy (x being 1 and y being 2).

- **Rotating a conic or quadric surface to its principal axes** is useful for eliminating the cross terms, thereby making it easier to analyze.

$$\begin{pmatrix} x' & y' \end{pmatrix} C^{-1} M C \begin{pmatrix} x' \\ y' \end{pmatrix} = K$$
$$\begin{pmatrix} x' & y' \end{pmatrix} D \begin{pmatrix} x' \\ y' \end{pmatrix} = K$$

  where

- M is the matrix of the coefficients of the conic or quadratic equation,
- D is the diagonal matrix of the eigenvalues of M,
- C is the matrix of the eigenvectors of M.

Note that if M is symmetric, then C is orthogonal, i.e., $C^{-1} = C^T$.

- **Multi-mass and -spring systems**: Consider a system of 2 masses m_1 and m_2 connected by 3 springs of spring constants k_1, k_2, and k_3. The system is one-dimensional and is in the form spring-mass-spring-mass-spring, in which the first and last springs are connected to fixed walls on the left and right, respectively. The objective is to find the equations of motion. The general steps are:

  1. Recall that a Hookian spring force is given by
  $$\mathbf{F} = -k\mathbf{x}$$
  where k is the spring constant and $\mathbf{x}$ is the displacement of the spring from its equilibrium position.
  2. Recall that
  $$\mathbf{F} = -\nabla U$$
  where U is the potential energy of the system. In one dimension:
  $$F = -\frac{\partial U}{\partial x}.$$
  3. If x_1 and x_2 are the displacements of the first and second masses from their equilibrium positions, respectively, then the middle spring's compression/extension is given by $x_2 - x_1$.
  4. Solve for U in terms of k_1, k_2, k_3, x_1, and x_2.
  5. Recall that
  $$-\frac{\partial U}{\partial x} = F = ma = m\ddot{x}.$$
  6. Since this involves haromic motion, assume the form
  $$x_n(t) = A_n e^{i\omega t},$$
  so
  $$\ddot{x}_n = -\omega^2 x_n.$$
  7. Designate
  $$\lambda_n = \frac{m\omega_n^2}{k},$$
  where, in fact, λ is the eigenvalue of the linear system.

8. Solve for the eigenvalues and eigenvectors of the system. The eigenvalues give the **characteristic frequencies** (when solving for ω_n) of the system, and the eigenvectors give the **characeristic modes of vibration** of the system. For this particular example, if both elements in an eigenvector are of the same sign, then the mode of vibration would be $\rightarrow\rightarrow$ and $\leftarrow\leftarrow$. If the elements are of opposite signs, then the mode of vibration at the corresponding frequency would be $\rightarrow\leftarrow$ and $\leftarrow\rightarrow$.

2 Vector Analysis

2.1 Scalar Fields

A scalar field $f(\mathbf{x})$ is a function that maps each point $\mathbf{x} = x_1\hat{x}_1 + x_2\hat{x}_2 + \cdots + x_n\hat{x}_n$ to a scalar: $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Some examples include temperature and pressure as a function of position.

2.2 Vector Fields

A vector field $\mathbf{F}(\mathbf{x}) = F_1(\mathbf{x})\hat{x}_1 + F_2(\mathbf{x})\hat{x}_2 + \cdots + F_m(\mathbf{x})\hat{x}_m$ is a function that maps each point $\mathbf{x} = x_1\hat{x}_1 + x_2\hat{x}_2 + \cdots + x_n\hat{x}_n$ to a vector: $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Note that each $F_i(\mathbf{x})$ is a scalar field. Some examples include velocity and acceleration as a function of position.

2.3 Dot Product

Also known as the scalar product, the dot product takes two vectors and returns a scalar.

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= |\mathbf{A}|\mathbf{B}| \cos\theta \\ &= A_x B_x + A_y B_y + A_z B_z\end{aligned}$$

Properties:

- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- $\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2$
- $\mathbf{A} \cdot \mathbf{B} = 0 \iff \mathbf{A} \perp \mathbf{B}$

Note that the dot product is *not* associative.

2.4 Projections

The scalar projection of $\mathbf{A}$ onto $\mathbf{B}$ is given by

$$\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = |\mathbf{A}| \cos\theta$$

The vector projection of $\mathbf{A}$ onto $\mathbf{B}$ is given by

$$\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|^2}\mathbf{B} = (\mathbf{A} \cdot \mathbf{u})\mathbf{u}$$

2.5 Gram-Schmidt Orthonormalization

Given any two vectors $\mathbf{A}$ and $\mathbf{B}$, we can find an orthonormal basis $\mathbf{v_1}$ and $\mathbf{v_2}$ for the plane spanned by $\mathbf{A}$ and $\mathbf{B}$ as

$$\mathbf{v_1} = \frac{\mathbf{A}}{|\mathbf{A}|} = \mathbf{u}$$
$$\mathbf{v_2} = \frac{\mathbf{B} - (\mathbf{B} \cdot \mathbf{u})\mathbf{u}}{|\mathbf{B} - (\mathbf{B} \cdot \mathbf{u})\mathbf{u}|}$$

2.6 Equation of a Plane

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r_0}) = 0$$
$$n_x(x - x_0) + n_y(y - y_0) + n_z(z - z_0) = 0$$

where

- $\mathbf{r} = (x, y, z)$ is a vector to any point on the plane,
- $\mathbf{n}$ is the normal vector to the plane, and
- $\mathbf{r_0} = (x_0, y_0, z_0)$ is a known point on the plane.

2.7 Cross Product

Also known as the vector product, the cross product takes two vectors and returns a vector.

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (|\mathbf{A}||\mathbf{B}| \sin\theta)\mathbf{u} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}\end{aligned}$$

where $\mathbf{u}$ is the unit vector perpendicular to both $\mathbf{A}$ and $\mathbf{B}$. Use the **right-hand rule** to determine the correct direction of $\mathbf{u}$. Geometrically, the magnitude of the cross product represents the area of the parallelogram created by $\mathbf{A}$ and $\mathbf{B}$.

Properties:

- Anti-commutative: $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$
- $\mathbf{A} \times \mathbf{B} = \mathbf{0} \iff \mathbf{A} \parallel \mathbf{B}$
  - Thus, $\mathbf{A} \times \mathbf{A} = \mathbf{0}$.

2.8 Equation of a 3D Line

$$\begin{aligned} \mathbf{r} &= \mathbf{r}_0 + t\mathbf{u} \\ \mathbf{r} \times \mathbf{u} &= \mathbf{r}_0 \times \mathbf{u} \end{aligned}$$

where

- $\mathbf{r} = (x, y, z)$ is a vector to any point on the line,
- $\mathbf{r}_0 = (x_0, y_0, z_0)$ is a known point on the line,
- $\mathbf{u}$ is a unit vector in the direction of the line, and
- t is a scalar parameter.

2.9 Triple Scalar Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

Performing an even number of swaps on the determinant results in the same answer (which is equivalent to performing **cyclic permutations** such as $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$), whereas performing an odd number of swaps simply results in an answer of the opposite sign.

Geometrically, the triple scalar product represents the volume of a parallelopiped formed by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. More specifically, any two vectors can be used to form the base of the parallelopiped (e.g., $\mathbf{A}$ and $\mathbf{B}$), and the height is the third vector's ($\mathbf{C}$) projection onto the normal vector of the base ($\mathbf{A} \times \mathbf{B}$). Furthermore, $\frac{1}{6}$ of the volume of the parallelopiped is the volume of the tetrahedron formed by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$.

2.10 Triple Vector Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

To remember this formula (beyond the mnemonic BAC-CAB), the right hand side is a linear combination of the vectors in the parentheses on the left hand side. The coefficients are the outside vector dotted with each of the inside vectors. Both terms have $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ exactly once. Finally, the positive term on the right hand side belongs to the vector in the middle on the left hand side ($\mathbf{B}$ in this case).

$\mathbf{B} \times \mathbf{C}$ is perpendicular to the plane spanned by $\mathbf{B}$ and $\mathbf{C}$, but $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is back on that plane.

2.11 Polar Coordinates

In rectangular coordinates, the basis vectors $\mathbf{i}$ and $\mathbf{j}$ are constant for all points, but in polar coordinates, the basis vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ are *not* constant.

$$\begin{aligned}
\mathbf{e}_r &= \cos\theta\ \mathbf{i} + \sin\theta\ \mathbf{j} \\
\mathbf{e}_\theta &= -\sin\theta\ \mathbf{i} + \cos\theta\ \mathbf{j} \\
\frac{d\mathbf{e}_r}{dt} &= \mathbf{e}_\theta \frac{d\theta}{dt} \\
\frac{d\mathbf{e}_\theta}{dt} &= -\mathbf{e}_r \frac{d\theta}{dt}
\end{aligned}$$

To find the derivative of a vector in polar coordinates, the chain rule must be remembered. In rectangular coordinates, the derivative of a basis vector is always zero, leading to the result being the simple derivative of the components, but this is *not* the case in polar coordinates.

2.12 Spherical Coordinates

$$\begin{aligned}
\mathbf{e}_\rho &= \cos\theta\sin\phi\ \mathbf{i} + \sin\theta\sin\phi\ \mathbf{j} + \cos\phi\ \mathbf{k} \\
\mathbf{e}_\theta &= -\sin\theta\ \mathbf{i} + \cos\theta\ \mathbf{j} \\
\mathbf{e}_\phi &= \cos\theta\cos\phi\ \mathbf{i} + \sin\theta\cos\phi\ \mathbf{j} - \sin\phi\ \mathbf{k}
\end{aligned}$$

Note that θ and ϕ can be swapped, but the convention is the class is to assign θ to the xy-plane and ϕ to the z-axis.

2.13 Gradient

The gradient represents the direction and magnitude of the greatest rate of increase of a scalar field. The vector opposite of the gradient is the direction of the greatest rate of decrease. The vectors perpendicular to the gradient represent no change in the scalar field.

In rectangular coordinates:

$$\boldsymbol{\nabla} f = \text{grad } f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

where f is a scalar field. Note that $\boldsymbol{\nabla} f$

- is a vector field.
- points in the direction of the greatest rate of increase of f.
- is perpendicular to the surface $f = c$ for any constant c.

In cylindrical coordinates:

$$\boldsymbol{\nabla} f = \text{grad } f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z$$

In spherical coordinates:

$$\boldsymbol{\nabla} f = \text{grad } f = \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho \sin\phi}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\mathbf{e}_\phi$$

2.14 Directional Derivative

$$\frac{df}{ds} = \boldsymbol{\nabla} f \cdot \mathbf{u}$$

where $\mathbf{u}$ is the unit vector in the direction of differentiation.

Special cases in rectangular coordinates:

- $\mathbf{u} = \mathbf{i} \implies \frac{df}{ds} = \frac{\partial f}{\partial x}$
- $\mathbf{u} = \mathbf{j} \implies \frac{df}{ds} = \frac{\partial f}{\partial y}$
- $\mathbf{u} = \mathbf{k} \implies \frac{df}{ds} = \frac{\partial f}{\partial z}$

These are simply the coefficients in front of the basis vectors from the definition of the gradient. The special cases for cylindrical and spherical coordinates can be found similarly.

2.15 Divergence

In rectangular coordinates:

$$\boldsymbol{\nabla} \cdot \mathbf{V} = \text{div } \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

where $\mathbf{V}$ is a vector field. Note that $\boldsymbol{\nabla} \cdot \mathbf{V}$ is a scalar field.

In cylindrical coordinates:

$$\boldsymbol{\nabla} \cdot \mathbf{V} = \frac{1}{r}\frac{\partial}{\partial r}(rV_r) + \frac{1}{r}\frac{\partial}{\partial \theta}V_\theta + \frac{\partial}{\partial z}V_z$$

In spherical coordinates:

$$\boldsymbol{\nabla} \cdot \mathbf{V} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 V_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(V_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial V_\phi}{\partial \phi}$$

2.16 Curl

In rectangular coordinates:

$$\begin{aligned}
\boldsymbol{\nabla} \times \mathbf{V} &= \text{curl } \mathbf{V} \\
&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \\
&= \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right)\mathbf{i} + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right)\mathbf{j} + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)\mathbf{k}
\end{aligned}$$

where $\mathbf{V}$ is a vector field. Note that $\boldsymbol{\nabla} \times \mathbf{V}$ is a vector field.

2.17 Laplacian

In rectangular coordinates:

$$\begin{aligned}
\nabla^2 f = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f) &= \text{div grad } f \\
&= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}
\end{aligned}$$

In cylindrical coordinates:

$$\nabla^2 f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}$$

In spherical coordinates:

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$

2.18 Vector Operator Properties

- A field is **irrotational** iff $\boldsymbol{\nabla} \times \mathbf{F} = \mathbf{0}$
  - Gradient fields are irrotational: $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0}$
- A field is **incompressible** (or **solenoidal**) iff $\boldsymbol{\nabla} \cdot \mathbf{F} = 0$
  - $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{F}) = 0$
- Linearity:
  - $\boldsymbol{\nabla}(f + g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g$ and $\boldsymbol{\nabla}(cf) = c\boldsymbol{\nabla} f$
  - $\boldsymbol{\nabla} \cdot (\mathbf{F} + \mathbf{G}) = \boldsymbol{\nabla} \cdot \mathbf{F} + \boldsymbol{\nabla} \cdot \mathbf{G}$ and $\boldsymbol{\nabla} \cdot (c\mathbf{F}) = c(\boldsymbol{\nabla} \cdot \mathbf{F})$
  - $\boldsymbol{\nabla} \times (\mathbf{F} + \mathbf{G}) = \boldsymbol{\nabla} \times \mathbf{F} + \boldsymbol{\nabla} \times \mathbf{G}$ and $\boldsymbol{\nabla} \times (c\mathbf{F}) = c(\boldsymbol{\nabla} \times \mathbf{F})$
- Product rule:
  - $\boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla} g + g\boldsymbol{\nabla} f$
  - $\boldsymbol{\nabla} \cdot (f\mathbf{F}) = f(\boldsymbol{\nabla} \cdot \mathbf{F}) + \mathbf{F} \cdot (\boldsymbol{\nabla} f)$
  - $\boldsymbol{\nabla} \times (f\mathbf{F}) = f(\boldsymbol{\nabla} \times \mathbf{F}) + (\boldsymbol{\nabla} f) \times \mathbf{F}$
- Gradient quotient rule:

$$\boldsymbol{\nabla}\left(\frac{f}{g}\right) = \frac{g\boldsymbol{\nabla} f - f\boldsymbol{\nabla} g}{g^2}$$

- $\boldsymbol{\nabla} \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\boldsymbol{\nabla} \times \mathbf{F}) - \mathbf{F} \cdot (\boldsymbol{\nabla} \times \mathbf{G})$
- $\nabla^2(fg) = f\ \nabla^2 g + g\ \nabla^2 f + 2(\boldsymbol{\nabla} f \cdot \boldsymbol{\nabla} g)$
- Using properties above: $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f \times \boldsymbol{\nabla} g) = 0$
- $\boldsymbol{\nabla} \cdot (f\boldsymbol{\nabla} g - g\boldsymbol{\nabla} f) = f\ \nabla^2 g - g\ \nabla^2 f$
- Using the triple vector product: $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{F}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{F}) - \nabla^2\mathbf{F}$
  - Note that this property also defines $\nabla^2\mathbf{F}$.

2.19 Unit Tangent and Normal Vectors

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \mathbf{r}'(s) = \frac{x'(t)\mathbf{i} + y'(t)\mathbf{j}}{\sqrt{x'(t)^2 + y'(t)^2}}$$

$$\mathbf{N} = \frac{y'(t)\mathbf{i} - x'(t)\mathbf{j}}{\sqrt{x'(t)^2 + y'(t)^2}}$$

where

- $\mathbf{r}(t) : \mathbb{R}^1 \to \mathbb{R}^n$ is a parameterization of a curve C,
- $x'(t)dt = dx$ and $y'(t)dt = dy$, and
- $\mathbf{N}$ points outward from the region enclosed by C.

2.20 Line Integrals and Circulation

For a scalar field f in 1D:

$$\int_C f\ ds = \int_a^b f\ dt$$

where C is the set of all points $t \in [a, b]$, and ds is a differential of arc length.

For a scalar field f in 2D and beyond:

$$\int_C f\ ds = \int_a^b f(\mathbf{r}(t))\ |\mathbf{r}'(t)|\ dt = \int_{-C} f\ ds$$

where $ds = |\mathbf{r}'(t)|\ dt$ and the direction of integration does not matter.

For a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ in 2D:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\ ds = \int_C P\ dx + Q\ dy = -\int_{-C} \mathbf{F} \cdot d\mathbf{r}$$

Note that the direction of integration matters for vector fields, unlike scalar fields. Also, $\mathbf{F} \cdot \mathbf{T} = |\mathbf{F}| \cos\theta$ represents the amount of $\mathbf{F}$ in the direction of C. Thus, the above integral represents the **circulation** of $\mathbf{F}$ along C, and $\mathbf{F} \cdot \mathbf{T}$ is the circulation or flow per unit length along C.

2.21 Flux

$$\oint_C \mathbf{F} \cdot d\mathbf{n} = \oint_C \mathbf{F}(\mathbf{r}(t)) \cdot (dy\mathbf{i} - dx\mathbf{j}) = \oint_C P\ dy - Q\ dx$$

where $\mathbf{dn} = \mathbf{N}\ ds$ pertains to a region enclosed by C.

2.22 Conservative Vector Fields

$$\oint_C \boldsymbol{\nabla} f \cdot d\mathbf{r} = 0$$
$$\int_C \boldsymbol{\nabla} f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

$\boldsymbol{\nabla} f$ is called a gradient field and is **conservative**, which means that the path taken does not matter. Note the difference between the two integrals. The first is a closed curve, while the second is an open curve. A physical example is potential difference: it is zero around a closed curve, and it is simply a difference between the values at the two endpoints for an open curve.

- Torque??

- Line integrals pg. 320, conservative vector fields pg. 322 and curl, gradient
- pg. 323 conservative -¿ potential
- pg. 325 vecAdvecA = AdA
- pg. 326 exact diff (and mixed partial equal) when conservative
- pg. 330 Green's Theorem
- pg. 332 tanget and outward normal vec
- pg. 332 9.13 9.14 divergence theorem
- pg. 333 9.18 9.19 Stokes' Theorem